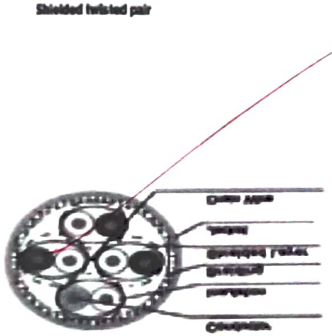


- UTP



Shielded twisted pair

- SSTP



- SSTP



Expt No: 2

Date: 10/07/2022

# STUDY OF DIFFERENT TYPES OF NETWORK CABELS

**Aim:**

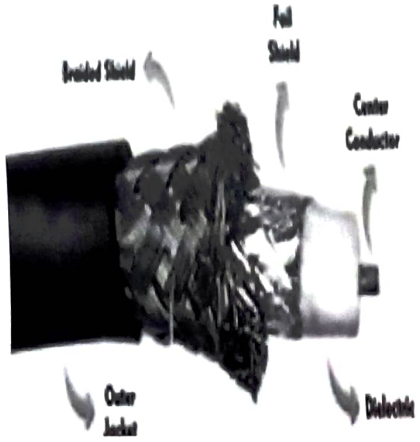
To Study and understand different types of network cables.

To make your own ethernet Cross over Cables / Straight Cables.

**Different Types of Network Cable:**

1. Unshielded Twisted pair Cable.
2. Shielded Twisted pair Cable.
3. Coaxial Cable.
4. Fibre optic Cable.

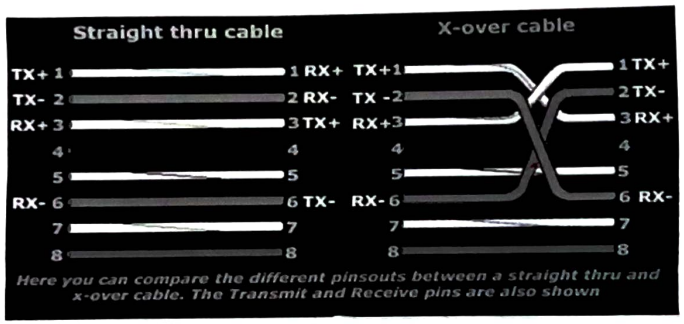
| Cable Type | Category | Maximum Data Transmission | Advantages / Disadvantages | Applications |
|---|---|---|---|---|
| UTP | Category 3 | 10 bps | **Advantages:-**<br>• Cheaper in cost<br>• easy to install as they have Small Overall diameter<br>**Disadvantages:**<br>• More prone to EMI | 10 Base -T ethernet |
| | Category 5 | upto 100 mbps | | Fast ethernet<br>Gigabit Ethernet |
| | Category 5e | 1 Gbps | | |
| STP | Category 6, 6a | 10 Gbps | **Advantages:**<br>• Shielded<br>• Faster than UTP<br>• Less prone to noise<br>**Disadvantages:**<br>• Expensive<br>• Greater installation process | • Gigabit ethernet<br>• Widely used in data center |
| SSTP | Category 7 | 10 Gbps | | • Gigabit Ethernet,<br>10G Ethernet. |

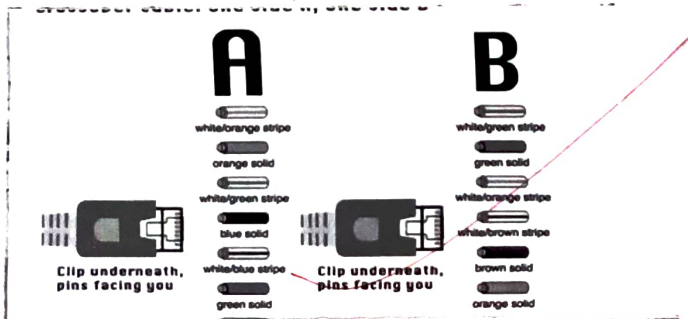


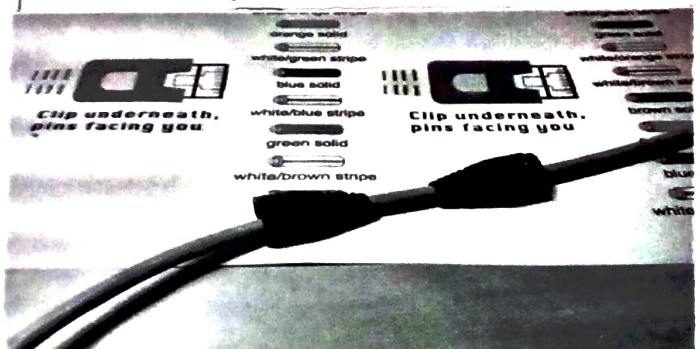

Coaxial Cable



fibre optics cable.




| | | | | |
|---|---|---|---|---|
| Coaxial Cable | RG-6<br>RG-59<br>RG-11 | 10-100 Mbps | Advantages :-<br>• High bandwidth<br>• Immune to interface<br>• versatile<br>Disadvantages :-<br>• cost<br>• Size is bulky | • Speed of Signal is 500 meter.<br>• High Speed internet connection |
| Fibre optic Cable | Singlemode<br>Multimode | 100Gbps | Advantages:<br>• High Speed.<br>• High bandwidth.<br>• High Security.<br>Disadvantages:<br>• Expensive | • Maximum distance of fibre optic cable is around 100 meter. |

Make Your Own Ethernet Cross-Over Cable | Straight Cable:

Tools and parts needed:-

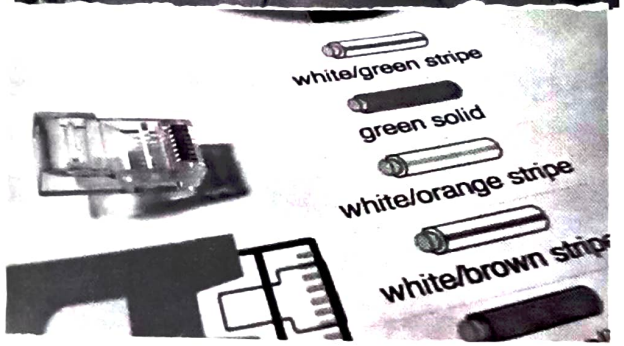
- Ether Cabling. Cat 5e is certified for gigabit support. but Cat 5. Cabling works as well.
- A Crimping tool.
- Two RJ45 plugs.
- optional two plug Shields.

Step 1:

To Start Construction of the device, begin by threading Shields on to the Cable.

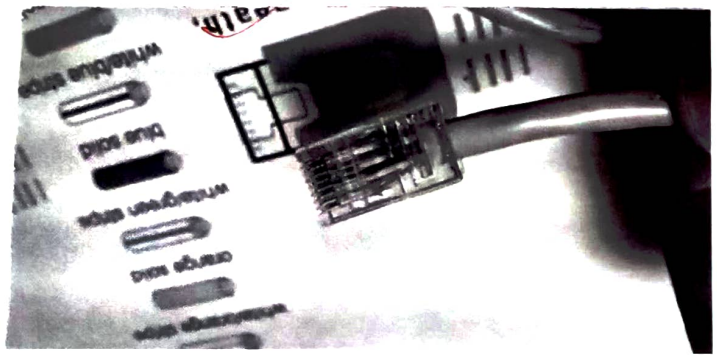
Step 2: Next Strip approximately 1.5 cm of the Cable Sheilding from both ends.









Step 3: ~~Step~~ After you will need to untangle the wires, there should be four twisted pairs.

Step 4: Once the order is correct, bunch them to gather in a line and if there are any that stick out farther than other snip them back to create an even level. The different aspect is placing the RJ45 plug without messing up the order.

Step 5: Next push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shieldi Simply snip the cables back a little more.

Step 6: After the wires are securely sitting inside the plug, inse it into the crimping tool and push down.

Step 7: Lastly, repeat for the other end using diagram B (to mak crossover cables) using diagram A (to make a straight through cable)

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Result:

Thus, different types of network cables were studied successfully and straight through and cross over ethernet cables were prepared and tested successfully.